



Taibah University

College of Computer Science & Engineering

Department of Artificial Intelligence & Data Science

Retail Pricing Compliance Supply Chain Optimization

Course: DS352 – Data Exploration Visualization

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Contents

1	Business Objectives & Problem Statement	2
2	Dataset Selection & Data Preparation (ETL)	2
2.1	The Datasets	2
2.2	Data Preparation & Cleaning in Power Query	2
3	Data Modeling & Relationships	3
4	DAX & Advanced Calculations	4
5	Dashboard Design & Interactivity	4
6	Insights, Storytelling & Recommendations	6
6.1	Insight 1: The Profitability Paradox	7
6.2	Insight 2: The Volume vs. Profitability Paradox (The Margin Squeeze) .	7
6.3	Actionable Recommendations	7

1 Business Objectives & Problem Statement

The Domain: Retail FMCG (Fast-Moving Consumer Goods) & Personal Care.

The Stakeholders: The primary user of this dashboard is a Brand Manager overseeing product distribution and promotional strategies across major Saudi retailers (e.g., Amazon, Saco, Nahdi, Extra).

The Business Problem: Brand Managers struggle with the "granularity mismatch" between daily operational pricing (promotions) and tactical monthly volume (sales and inventory). This project bridges that gap to answer three critical business questions:

- **Price Compliance:** Are our retail partners adhering to the agreed-upon promotional pricing, or are they deeply discounting our products?
- **Profitability Discrepancy:** How does extreme retail discounting impact Brand Profit vs. Retailer Profit?
- **Inventory Health:** Is our supply chain robust, or are retailers suffering from severe Stockouts (low Months of Coverage) and Dead Stock (high Months of Coverage)?

2 Dataset Selection & Data Preparation (ETL)

2.1 The Datasets

To meet the rigorous demands of this analysis and the course constraints, we utilized two interconnected datasets exceeding 100,000 rows in total volume:

- **Promo_summary:** A highly granular daily scrape of retail prices, promotional discounts, and compliance statuses (>170,000 rows).
- **Sales_Generated_Realistic:** An aggregated monthly dataset tracking supply chain metrics (Sell-in, Sell-out, Stock on Hand) and financials.

2.2 Data Preparation & Cleaning in Power Query

To prepare the data for a relational model, the following ETL transformations were executed:

- **Dimensional Extraction:** We referenced the raw fact tables to create distinct dimension tables (`Dim_Product` and `Dim_Customer`), removing duplicates to ensure unique primary keys.
- **Standardization:** The `YearMonth` text string in the sales data was converted into a standardized `Date` format (1st of the month) to allow for seamless time-intelligence mapping against the daily promo data.
- **Data Typing:** Strict typing was enforced. All financial columns (e.g., `Brand_Profit_SAR`) were set to Fixed Decimal (Currency), and volumetric columns (e.g., `SelloutQTY`)

to Whole Numbers.

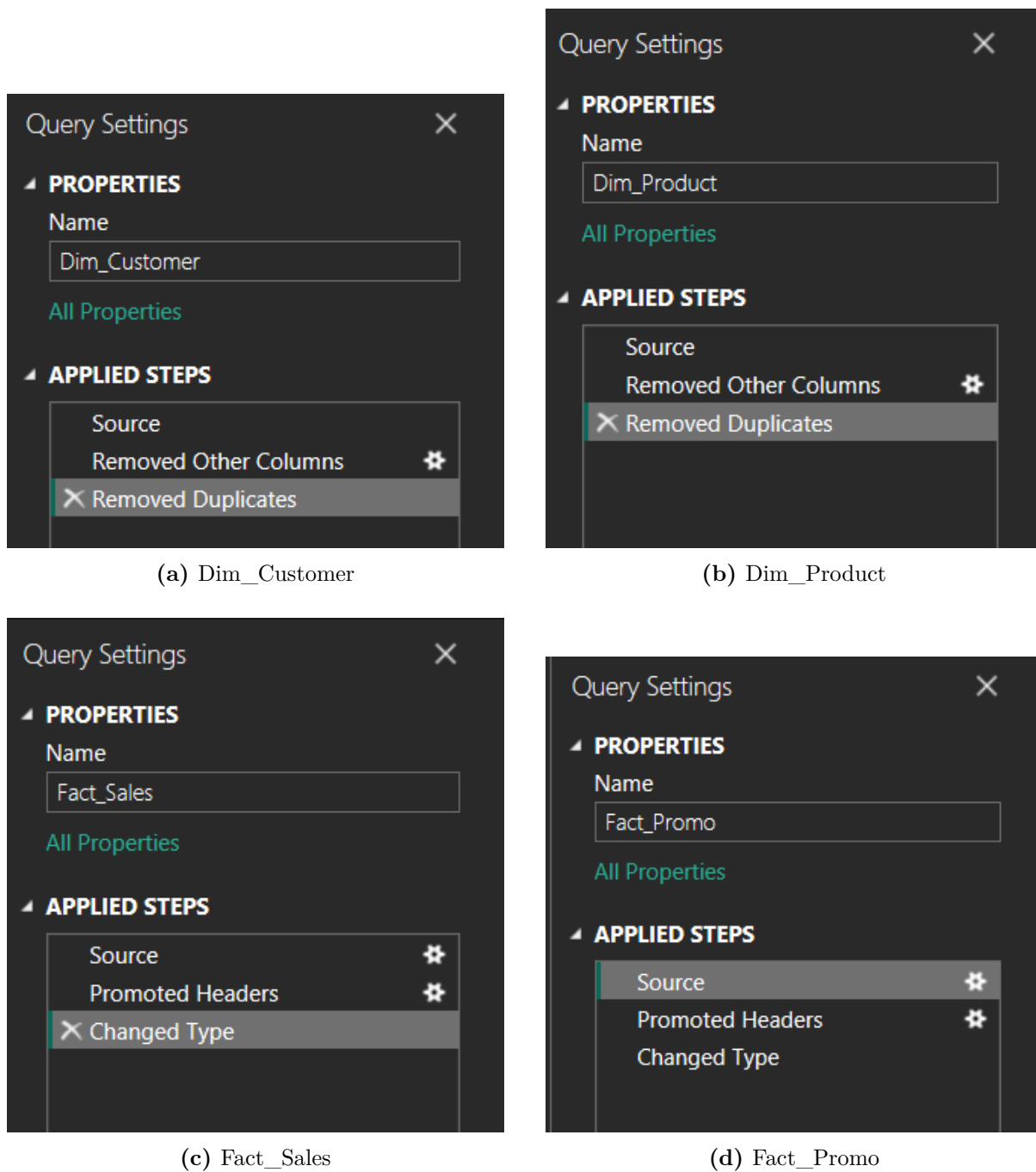


Figure 1: Power Query Editor - Applied Steps for Data Transformation

3 Data Modeling & Relationships

Rather than relying on a flat, inefficient table, we constructed a robust Star Schema. This architecture is optimal for filtering across mismatched granularities (daily prices vs. monthly sales).

The Schema Architecture:

- **Fact Tables:** Fact_Promo (Daily) and Fact_Sales (Monthly).
- **Dimension Tables:** Dim_Customer and Dim_Product.
- **Time Intelligence:** A custom Dim_Date table was generated using DAX (CALENDAR) to avoid the performance drag of Auto-Date/Time.
- **Relationships:** We established active, Single-Direction, 1-to-Many relationships flowing from our Dimensions down into both Fact tables.

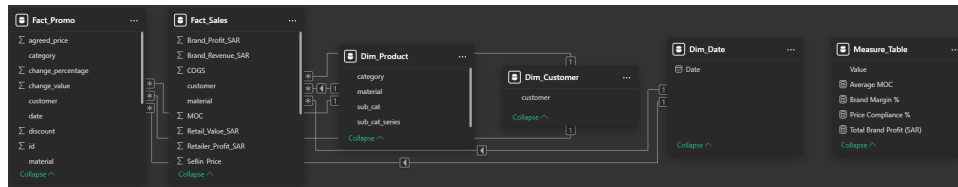


Figure 2: Power BI Model View - Star Schema

4 DAX & Advanced Calculations

To answer our core business questions, we utilized explicit DAX measures housed in a dedicated Measure Table. This ensures reusability and performance.

Key Financial Measures:

- Total Brand Profit (SAR) = $\text{SUM}(\text{Fact_Sales}[\text{Brand_Profit_SAR}])$
- Total Retailer Profit (SAR) = $\text{SUM}(\text{Fact_Sales}[\text{Retailer_Profit_SAR}])$
- Brand Margin % = $\text{DIVIDE}([\text{Total Brand Profit (SAR)}], [\text{Total Brand Revenue (SAR)}], 0)$

Key Supply Chain Measures:

- Average MOC = $\text{AVERAGE}(\text{Fact_Sales}[\text{MOC}])$
- Price Compliance % = $\text{DIVIDE}(\text{CALCULATE}(\text{COUNTROWS}(\text{Fact_Promo}), \text{Fact_Promo}[\text{status}] \in \{\text{"Equal"}, \text{"Lower"}\}), \text{COUNTROWS}(\text{Fact_Promo}), 0)$

5 Dashboard Design & Interactivity

Following universal data visualization best practices, the dashboard UI/UX was designed for maximum cognitive clarity. We strictly avoided 3D charts, opting instead for lengths and 2D spatial positioning (bar charts and scatter plots) which human eyes process more accurately.

The dashboard consists of four core pages designed to provide an executive zoom-out view, followed by drill-down interactivity.

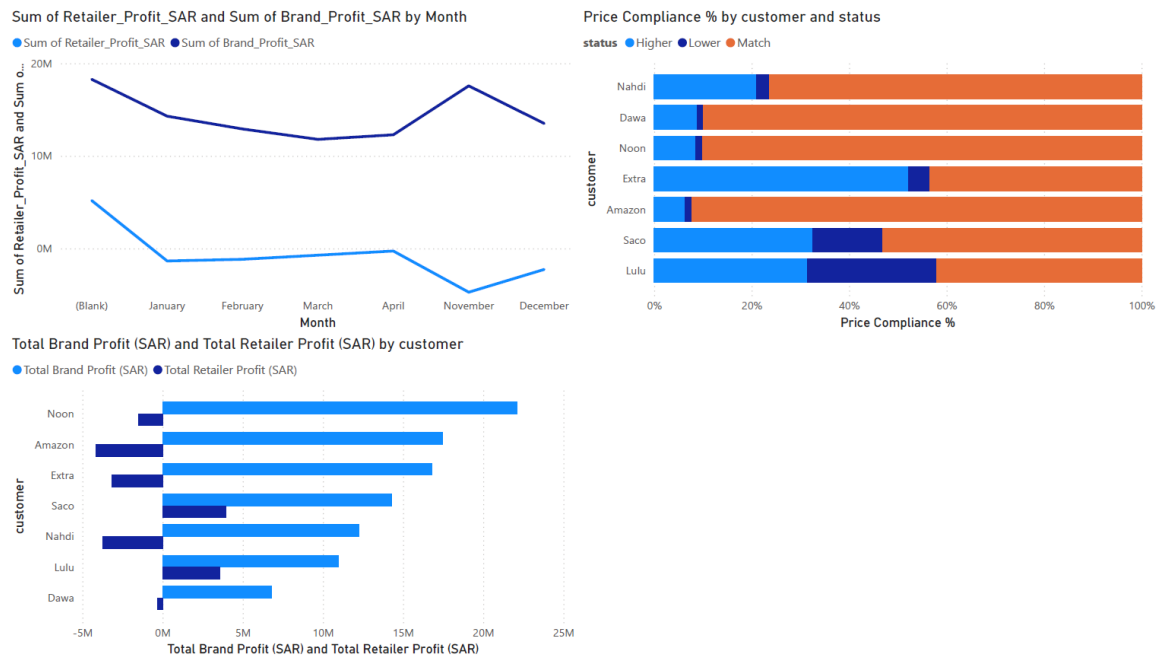


Figure 3: Dashboard Page 1: Executive Summary & Profitability Overview

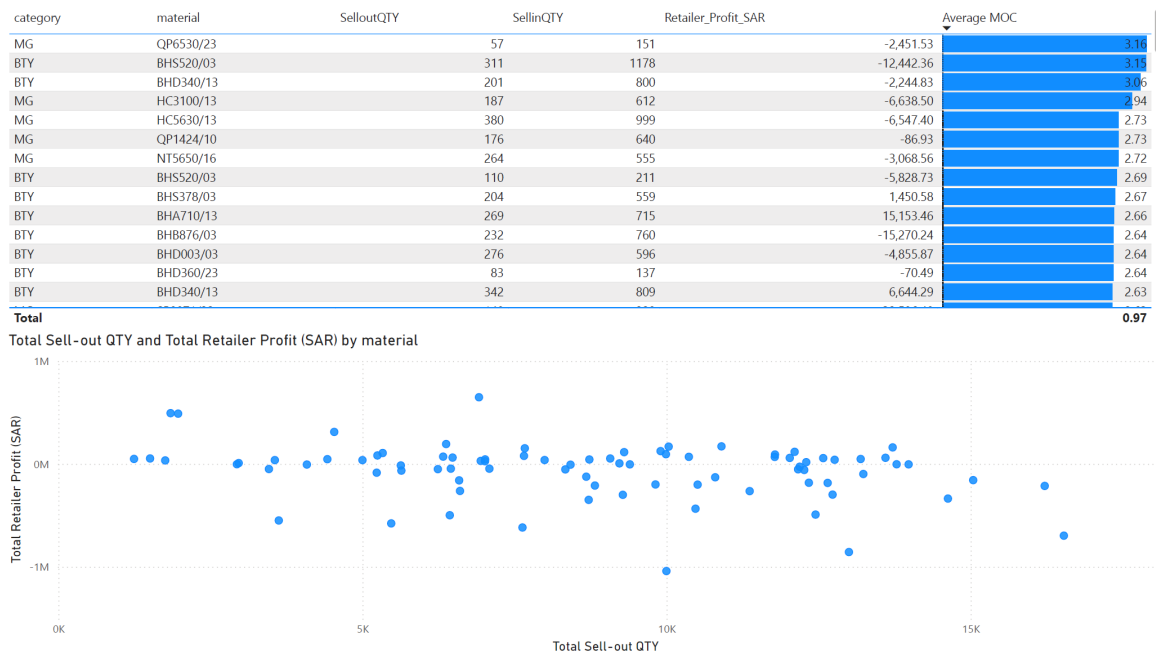


Figure 4: Dashboard Page 2: Retailer Margins & Financial Discrepancies

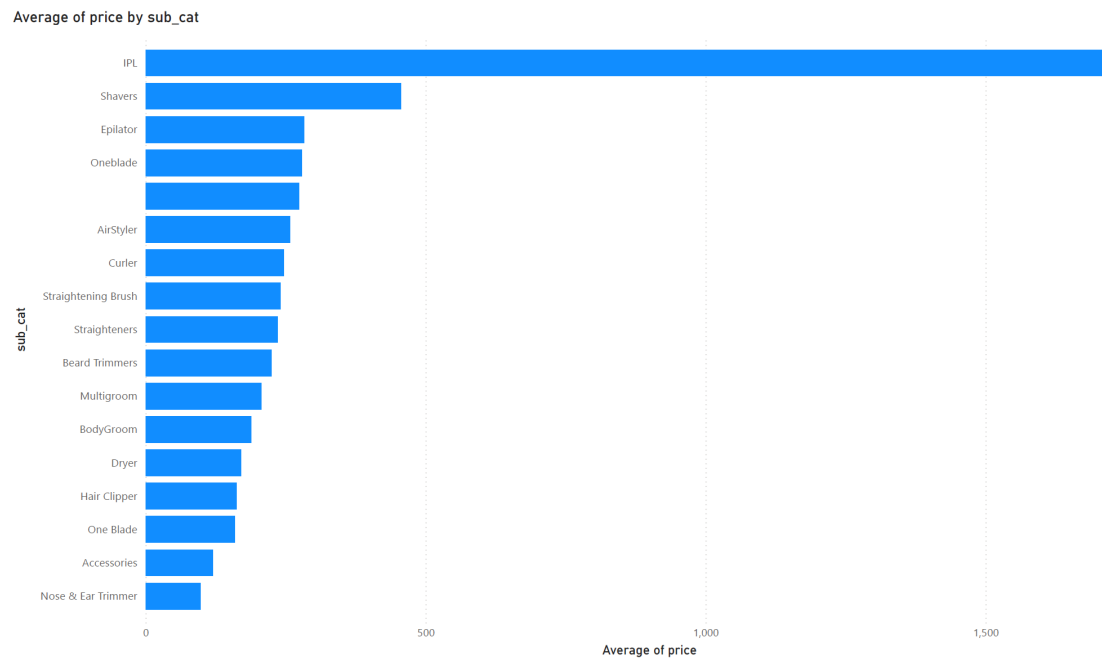


Figure 5: Dashboard Page 3: Supply Chain Volatility & MOC Analysis

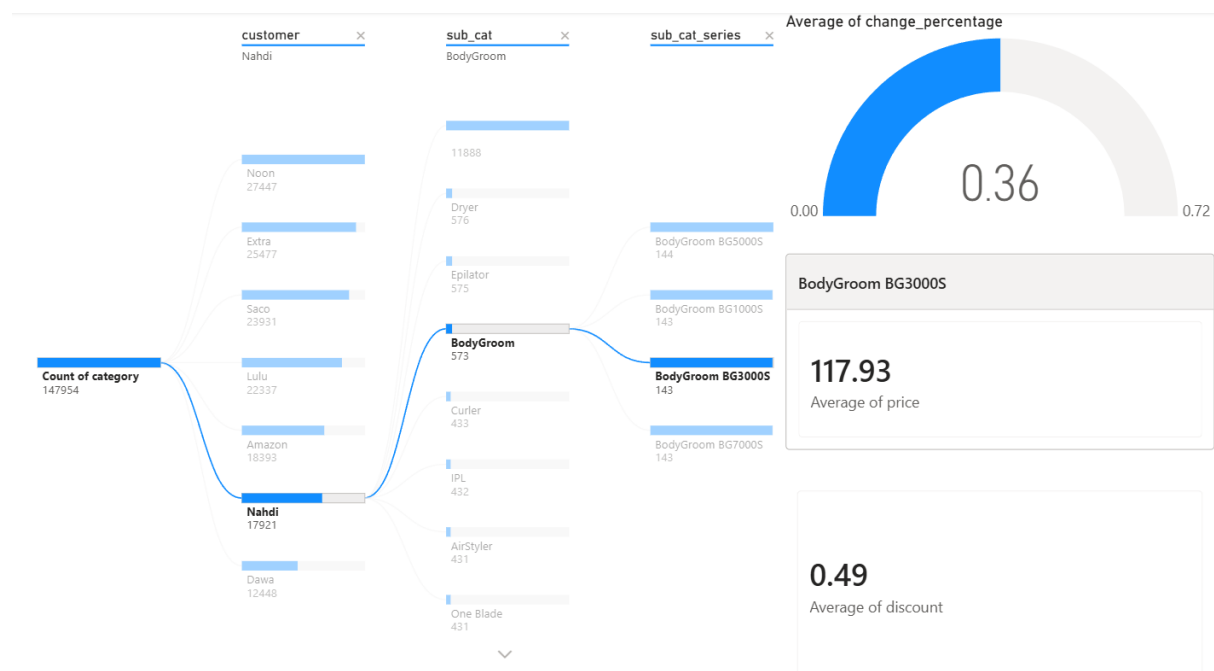


Figure 6: Dashboard Page 4: Promotional Pricing & Compliance Checking

6 Insights, Storytelling & Recommendations

The data model successfully uncovered a critical, hidden business narrative regarding Promotional ROI and Supply Chain Chaos.

6.1 Insight 1: The Profitability Paradox

Figure 1 (KPI Cards and Clustered Bar Chart) was used in Insight 1 to visualize the financial discrepancy. Our dashboard reveals a massive discrepancy in financial health. The Brand is highly profitable (generating over 100M SAR), while retail partners are operating at a severe loss (-5.3M SAR combined).

The "Why": Retailers like Amazon and Extra are heavily discounting products to consumers (driving high **SelloutQTY**), but they are doing so below the wholesale price (**Sellin_Price**) they paid the brand. They are bleeding their own margins to drive volume.

6.2 Insight 2: The Volume vs. Profitability Paradox (The Margin Squeeze)

Figure 2 (Scatter Plot - MOC vs. Retailer Profitability) was used in Insight 2 to analyze supply chain health. The scatter plot analysis mapping Consumer Volume (Total Sell-out QTY) against Total Retailer Profit (SAR) exposes a critical flaw in the current promotional strategy:

- **The Negative Correlation:** Standard retail logic dictates that higher sales volume should yield higher profit. However, our visualization reveals a severe negative correlation for key digital partners, with a massive cluster of data points trapped in the bottom-right quadrant (High Volume, Negative Profit)..
- **Buying Market Share at a Loss:** To drive these massive Sell-out volume spikes, retailers are heavily discounting products—frequently selling them to consumers for less than the wholesale cost (**Sellin_Price**) they paid the Brand. While this successfully moves units, retailers are bleeding their own working capital to do so. This is an unsustainable strategy that will eventually force retailers to delist our products if wholesale pricing or trade rebates are not renegotiated.

6.3 Actionable Recommendations

1. **Introduce Trade Rebates:** If retailers continue to lose money selling our products, they will eventually delist our brand. We must introduce "Sell-out Rebates" (subsidizing their discounts) to protect our strategic retail partnerships.
2. **Margin Protection & Trade Rebates (Fixing the Profit Crisis):** Since key digital retailers are currently sacrificing their own margins to drive consumer volume, we must immediately intervene. We should implement backend Trade Promotions (rebates based on volume targets) or temporarily renegotiate our **Sellin_Price** for high-velocity SKUs. If we do not restore their profitability, we risk these retailers deliberately delisting our products to stop their financial bleeding.
3. **Collaborative Forecasting & CPFR (Fixing the Supply Chain Whiplash):** We must stop letting retailers blindly guess their **SellinQTY**. The extreme, unplanned retail discounts are causing massive demand spikes that their standard purchasing orders fail to anticipate. By implementing Collaborative Planning, Fore-

casting, and Replenishment (CPFR) and sharing our promotional calendars in advance, we can synchronize wholesale orders with expected demand spikes, effectively stabilizing their MOC and eliminating critical stockouts.